

The figure displays ten quantum circuit diagrams, each representing a step in a quantum algorithm. The qubits are labeled as $q_4 \mapsto 39$, $q_0 \mapsto 52$, $q_3 \mapsto 53$, $q_2 \mapsto 54$, $q_1 \mapsto 55$, $\text{position} \mapsto 3$, and $\text{ancilla} \mapsto 2$. The circuits consist of various quantum gates, including R_z (rotation gates) and $\sqrt{X}$ (square root of X gates), with parameters such as $\pi/2$, $\pi/4$, $\pi/8$, $\pi/16$, $\pi/32$, $\pi/64$, $\pi/128$, $\pi/256$, $\pi/512$, $\pi/1024$, $\pi/2048$, $\pi/4096$, $\pi/8192$, $\pi/16384$, $\pi/32768$, $\pi/65536$, $\pi/131072$, $\pi/262144$, $\pi/524288$, $\pi/1048576$, $\pi/2097152$, $\pi/4194304$, $\pi/8388608$, $\pi/16777216$, $\pi/33554432$, $\pi/67108864$, $\pi/134217728$, $\pi/268435456$, $\pi/536870912$, $\pi/1073741824$, $\pi/2147483648$, $\pi/4294967296$, $\pi/8589934592$, $\pi/17179869184$, $\pi/34359738368$, $\pi/68719476736$, $\pi/137438953472$, $\pi/274877906944$, $\pi/549755813888$, $\pi/1099511627776$, $\pi/2199023255552$, $\pi/4398046511104$, $\pi/8796093022208$, $\pi/17592186044416$, $\pi/35184372088832$, $\pi/70368744177664$, $\pi/140737488355328$, $\pi/281474976710656$, $\pi/562949953421312$, $\pi/1125899906842624$, $\pi/2251799813685248$, $\pi/4503599627370496$, $\pi/9007199254740992$, $\pi/18014398509481984$, $\pi/36028797018963968$, $\pi/72057594037927936$, $\pi/144115188075855872$, $\pi/288230376151711744$, $\pi/576460752303423488$, $\pi/1152921504606846976$, $\pi/2305843009213693952$, $\pi/4611686018427387904$, $\pi/9223372036854775808$, $\pi/18446744073709551616$, $\pi/36893488147419103232$, $\pi/73786976294838206464$, $\pi/147573952589676412928$, $\pi/295147905179352825856$, $\pi/590295810358705651712$, $\pi/1180591620717411303424$, $\pi/2361183241434822606848$, $\pi/4722366482869645213696$, $\pi/9444732965739290427392$, $\pi/18889465931478580854784$, $\pi/37778931862957161709568$, $\pi/75557863725914323419136$, $\pi/151115727451828646838272$, $\pi/302231454903657293676544$, $\pi/604462909807314587353088$, $\pi/1208925819614629174706176$, $\pi/2417851639229258349412352$, $\pi/4835703278458516698824704$, $\pi/9671406556917033397649408$, $\pi/19342813113834066795298816$, $\pi/38685626227668133590597632$, $\pi/77371252455336267181195264$, $\pi/154742504910672534362390528$, $\pi/309485009821345068724781056$, $\pi/618970019642690137449562112$, $\pi/1237940039285380274899124224$, $\pi/2475880078570760549798248448$, $\pi/4951760157141521099596496896$, $\pi/9903520314283042199192993792$, $\pi/19807040628566084398385987584$, $\pi/39614081257132168796771975168$, $\pi/79228162514264337593543950336$, $\pi/158456325028528675187087900672$, $\pi/316912650057057350374175801344$, $\pi/633825300114114700748351602688$, $\pi/1267650600228229401496703205376$, $\pi/2535301200456458802993406410752$, $\pi/5070602400912917605986812821504$, $\pi/10141204801825835211973625643008$, $\pi/20282409603651670423947251286016$, $\pi/40564819207303340847894502572032$, $\pi/81129638414606681695789005144064$, $\pi/162259276829213363391578010288128$, $\pi/324518553658426726783156020576256$, $\pi/649037107316853453566312041152512$, $\pi/1298074214633706907132624082305024$, $\pi/2596148429267413814265248164610048$, $\pi/5192296858534827628530496329220096$, $\pi/10384593717069655257060992658440192$, $\pi/20769187434139310514121985316880384$, $\pi/41538374868278621028243970633760768$, $\pi/83076749736557242056487941267521536$, $\pi/166153499473114484112975882535043072$, $\pi/332306998946228968225951765070086144$, $\pi/664613997892457936451903530140172288$, $\pi/1329227995784915872903807060280344576$, $\pi/2658455991569831745807614120560689152$, $\pi/5316911983139663491615228241121378304$, $\pi/10633823966279326983230456482242756608$, $\pi/21267647932558653966460912964485513216$, $\pi/42535295865117307932921825928971026432$, $\pi/85070591730234615865843651857942052864$, $\pi/170141183460469231731687303715884105728$, $\pi/340282366920938463463374607431768211456$, $\pi/680564733841876926926749214863536422912$, $\pi/1361129467683753853853498429727072845824$, $\pi/2722258935367507707706996859454145691648$, $\pi/5444517870735015415413993718908291383296$, $\pi/10889035741470030830827987437816582766592$